Jaswanth

Forward Deployed Engineer · Agentic AI Systems · Full Stack Architect

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IEEE-PUBLISHED AUTHOR | AWS CERTIFIED SOLUTIONS ARCHITECT | AWS CERTIFIED MACHINE LEARNING | 12+ SHIPPED PRODUCTION SYSTEMS

SUMMARY

Forward Deployed Engineer and AI Systems builder with deep focus on autonomous agent orchestration, production LLM pipelines, and low-latency full-stack architecture. Track record of translating ambiguous enterprise requirements into reliable, deterministic software with strict safety boundaries and human-in-the-loop controls.

FEATURED AI ENGINEERING PROJECTS

OmniAgent Enterprise Orchestrator **2025 – 2026**

Multi-Agent System & Automated Tool Synthesis *Python, LangGraph, FastAPI, Redis*

- Architected asynchronous multi-agent coordination loop handling 100+ concurrent reasoning sessions with deterministic rollback and self-healing error recovery.
- Implemented dynamic tool schema extraction and sandboxed execution, reducing prompt token overhead by 38% and eliminating API execution failures.

Cognitive RAG Intelligence Engine **2025**

Enterprise Hybrid Vector & Knowledge Graph Pipeline *Qdrant, LlamaIndex, vLLM, PyTorch*

- Built sub-second hybrid retrieval pipeline combining dense vector embeddings with sparse BM25 and knowledge graph cross-referencing.
- Constructed real-time hallucination scoring and automated citation verification layer, achieving 99.4% factual precision in document Q&A benchmarks.

VisionOps Real-Time Multimodal Triage **2025**

Edge-Accelerated Vision-Language Inference *YOLO, ONNX Runtime, WebSockets, TypeScript*

- Engineered low-latency visual anomaly triage pipeline processing 60 FPS live video feeds at <45ms inference latency.
- Integrated automated incident summarization and multi-channel Slack/PagerDuty notification loops with verifiable audit logs.

TECHNICAL EXPERTISE

Agentic & LLMs:	LangGraph, AutoGen, CrewAI, DSPy, LlamaIndex, Tool Calling, Function Calling, Prompt Optimization
ML & Inference:	PyTorch, vLLM, ONNX Runtime, Hugging Face, Qdrant, ChromaDB, Fine-Tuning (LoRA/QLoRA)
Backend & Systems:	Python, TypeScript, FastAPI, Node.js, PostgreSQL, pgvector, Redis, Docker, Kubernetes, WebSockets
Cloud & DevOps:	AWS (Lambda, ECS, S3, SageMaker), GCP, CI/CD GitHub Actions, OpenTelemetry Tracing

PUBLICATIONS & CERTIFICATIONS

- **IEEE Published Paper**: Research in multimodal deep learning and agentic feedback architectures.
- **AWS Certified Solutions Architect – Associate**
- **AWS Certified Machine Learning – Specialty**